

EXPERIMENT 4

Impact of Principal Component Analysis on K-Nearest Neighbors Classifier Performance

1. Overview and Learning Objectives

This experiment investigates the computational and diagnostic impact of dimensionality reduction via Principal Component Analysis (PCA) when applied to distance-sensitive machine learning algorithms—specifically the K-Nearest Neighbors (KNN) classifier. Real-world tabular datasets frequently encompass mixed data types, sparse categorical distributions, and missing values. Preprocessing these data matrices through one-hot encoding substantially amplifies feature dimensionality, exposing distance-based classifiers to the curse of dimensionality. The primary objective is to build an end-to-end preprocessing pipeline, establish a baseline high-dimensional model, implement PCA from mathematical first principles in NumPy, and perform rigorous empirical comparisons with scikit-learn's implementation.

2. Exploratory Data Analysis (EDA) Findings

- **Missing Value Patterns:** In the Adult Census Income dataset, missing records are explicitly encoded as '?' rather than standard nulls. Inspection revealed that missing values are confined to three categorical attributes: workclass (approx. 5.63%), occupation (approx. 5.66%), and native-country (approx. 1.79%). Continuous numeric features (e.g., age, fnlwgt, hours-per-week) contain complete records with no null values.
- **Feature Distributions & Imbalance:** The target attribute (income) manifests a significant class imbalance with approximately 76% of instances earning $\leq 50K$ and 24% earning $> 50K$. Age distributions exhibit moderate right-skewness, peaking in the 25–45 range, with higher median ages for individuals earning $> 50K$. The hours-per-week feature exhibits an acute spike at the standard 40-hour work week, with higher earners exhibiting a broader, heavier distribution toward 45–60 hours.
- **Categorical Cardinality:** Categorical features contain varied cardinality. Transforming workclass, education, marital-status, occupation, relationship, race, sex, and native-country via one-hot encoding expands the input feature space from 14 raw variables to over 100 dimensions, inducing substantial feature sparsity.

3. Preprocessing Pipeline Architecture

To eliminate data leakage, a modular ColumnTransformer was structured and fitted solely on the training partition (80/20 train-test split, random_state=42, stratified across classes):

1. **Numerical Transformation Pipeline:** Missing entries imputed using the median (SimpleImputer(strategy='median')), followed by z-score standardization via StandardScaler. Standard scaling is essential so that distance metrics and variance calculations are not dominated by high-magnitude scales.
2. **Categorical Transformation Pipeline:** Missing entries imputed using the mode (SimpleImputer(strategy='most_frequent')), followed by OneHotEncoder(handle_unknown='ignore', sparse_output=False).

Total Transformed Dimensionality: The resulting preprocessed feature matrix `X_train_proc` expanded to 108 columns.

4. Experimental Results & Comparative Analysis

Three distinct KNN classification configurations ($k=5$ nearest neighbors) were evaluated: (1) Baseline KNN operating on the complete 108-dimensional space, (2) MyPCA + KNN utilizing custom NumPy PCA, and (3) `sklearn.PCA` + KNN utilizing scikit-learn's SVD-based PCA. The variance threshold was set to capture 95% of total variance.

Model Configuration	Features (d)	Accuracy	Fit Time (s)	Inference Time (s)
Model 1: Baseline KNN	108	83.21%	0.0382 s	6.8241 s
Model 2: MyPCA + KNN (Scratch)	44	83.47%	0.0164 s	2.7120 s
Model 3: sklearn.PCA + KNN	44	83.47%	0.0171 s	2.6984 s
Net Impact / Performance Gain	-59.3%	+0.26%	~2.3x Faster	~2.5x Faster

Discussion & Analytical Observations

1. Runtime Complexity and Inference Acceleration: K-Nearest Neighbors is an instance-based lazy learner that performs no explicit model parameter optimization during training; training time represents data indexing. Inference time, however, requires calculating Euclidean distances between each query test instance and all training instances: $O(N_{\text{test}} \cdot N_{\text{train}} \cdot d)$. By reducing dimensions from $d=108$ to $d=44$ (a 59.3% reduction in dimensionality), the distance computation load was slashed by nearly 60%, driving prediction latency from 6.82 seconds down to 2.70 seconds (a 2.52x speedup).

2. Accuracy Comparison: MyPCA vs. sklearn.PCA: The accuracy of MyPCA + KNN and `sklearn.PCA` + KNN was identical at 83.47%. Mathematically, MyPCA computes the eigendecomposition of the sample covariance matrix ($C = (1 / (n-1)) \cdot X^T X$) using `np.linalg.eigh`, whereas `sklearn.decomposition.PCA` computes the Singular Value Decomposition (SVD) directly on the centered data matrix ($X = U \Sigma V^T$). The eigenvectors obtained from covariance matrix diagonalization are mathematically equivalent to the right-singular vectors V from SVD. Furthermore, while individual eigenvectors are subject to sign ambiguity (arbitrary ± 1 factor), Euclidean distance is invariant to sign reflections along orthogonal coordinate axes ($||\pm x_i - (\pm x_j)||_2 = ||x_i - x_j||_2$). Consequently, the neighbor topologies and predictions remain strictly identical.

3. Mitigation of the Curse of Dimensionality: In addition to accelerating query speeds, PCA slightly improved classification accuracy from 83.21% to 83.47%. In the 108-dimensional one-hot encoded space, data becomes highly sparse, and pairwise distances concentrate around a narrow range (distance concentration phenomenon). By projecting the data onto the 44 principal directions accounting for 95% variance, redundant collinear dummy columns and noisy orthogonal variations were filtered, sharpening the margin between nearest neighbors.

5. Conclusion & Practical Learnings

This experiment demonstrated the indispensable utility of Principal Component Analysis as a feature extraction and dimension-reduction tool prior to distance-based classification. Key learnings include:

- ColumnTransformer encapsulation: Crucial for preventing feature leakage during scaling and imputation across cross-validation splits.
- Mathematical duality: Verified that covariance eigen-decomposition and data SVD yield identical projection geometries.
- Distance sensitivity: Confirmed that PCA effectively addresses the curse of dimensionality for KNN, yielding substantial inference speedups (2.5x) while preserving (and slightly enhancing) predictive accuracy.